

Chapter

Encapsulation

1. Which of the following access control must be used for the main method to be invoked by the JVM? SQP 2024

- (a) Private
- (b) Public
- (c) Protected
- (d) No access

2. Which of the following keywords helps access a class member without creating an object and without associating a variable with an object? COMP 2023

- (a) Private
- (b) Final
- (c) Static
- (d) Protected

3. Which of these is a non-access modifier in Java? COMP 2021

- (a) Public
- (b) Private
- (c) Final
- (d) Protected

4. Which access modifier allows variables to be accessible in all classes, including other packages? SQP 2021

- (a) Public
- (b) Private
- (c) Default
- (d) Protected

5. Which modifier restricts accessibility to the same class only? COMP 2023

- (a) Public
- (b) Private
- (c) Protected
- (d) Default

6. Define encapsulation. Why is it important? MAIN 2024

7. List the four access modifiers provided by Java. SQP 2023

8. What is the purpose of access modifiers? COMP 2021

Solution

1. (b) Public

The main method must be declared public so that the Java Virtual Machine (JVM) can access and invoke it as the entry point of the program.

2. (c) Static

The static keyword allows a class member (variable or method) to be accessed without creating an object of the class. Static members belong to the class itself rather than to any specific instance.

3. (c) Final

The final keyword in Java is a non-access modifier. It is used to make variables constant, prevent method overriding, and prevent class inheritance.

4. (a) Public

The public access modifier allows variables and methods to be accessed from any class, including those in other packages.

5. (b) Private

The private access modifier restricts access to variables and methods so that they can only be used within the same class where they are declared.

6. Encapsulation is the process of combining data (attributes) and methods (functions) into a single unit called a class. It is important because it protects the data from unauthorized access, ensures controlled access through methods, and helps reduce complexity by hiding the internal implementation.

7. The four access modifiers in Java are:

1. **Public**
2. **Protected**
3. **Private**
4. **Default** (also called Friendly access)

8. The purpose of access modifiers is to control the visibility and accessibility of class members (variables and methods), ensuring proper data protection and controlled access across classes and packages.